

Pharmaceutical Economic Data Hub

Introduction and Research Problem

The pharmaceutical pricing ecosystem in the United States is highly fragmented. Critical information about drug pricing, therapeutic equivalence, packaging, and regulatory approval exists across multiple public datasets. While each dataset provides valuable information independently, they are not designed to interoperate. Differences in schema design, naming conventions, identifiers, and update frequency make cross-dataset analysis difficult and error-prone.

Our project addresses this fragmentation by building a centralized source of truth database that aggregates and normalizes pharmaceutical economic data into a unified, queryable system. The main technical challenges include entity resolution across datasets, maintaining referential integrity across many-to-many relationships (drugs, NDCs, package sizes, pricing tiers), and designing a system that supports historical tracking and future analytical use cases.

Our minimum viable product is a fully normalized relational database hosted in Azure, supported by automated ingestion pipelines and an internal REST API.

Research Component: Price Analysis

In addition to building the infrastructure layer, we will conduct a comparative study of pharmaceutical price forecasting methodologies. Drug pricing exhibits characteristics similar to financial time series, including volatility, discontinuities, regulatory shocks, and structural breaks. However, unlike stock markets, pharmaceutical pricing is influenced by policy updates, patent expiration events, and therapeutic equivalence substitutions.

Existing approaches to forecasting economic time series include statistical models such as ARIMA and Prophet. There are also financial modeling techniques such as rolling-window regression, volatility clustering models, and anomaly-aware forecasting frameworks. In practice, pharmaceutical pricing analyses often rely on simple trend extrapolation or descriptive analytics rather than rigorous comparative forecasting evaluation.

Our research question is: Which class of forecasting models is most appropriate for pharmaceutical price volatility when applied to normalized, cross-source integrated pricing data? We will evaluate forecasting approaches using historical NADAC price data indexed by time, measuring performance under scenarios of stable pricing, gradual inflation, and regulatory shifts

Existing Approaches

In practice, pharmaceutical economic analysis is often performed using siloed datasets. Organizations may download NADAC or Orange Book files and run internal analyses without deeply integrating them. While this works for narrow tasks, it does not support reliable cross-source mapping or historical tracking.

Commercial vendors provide integrated drug databases, but these are proprietary, expensive, and not transparent in how they reconcile inconsistencies. From a research perspective, integrating heterogeneous datasets typically involves entity resolution techniques, schema normalization, and historical versioning strategies. However, few open systems attempt to integrate multiple major public pharmaceutical regulatory and pricing datasets simultaneously in a structured and extensible way.

Our project aims to bridge this gap by implementing data engineering principles in a practical, production-oriented system.

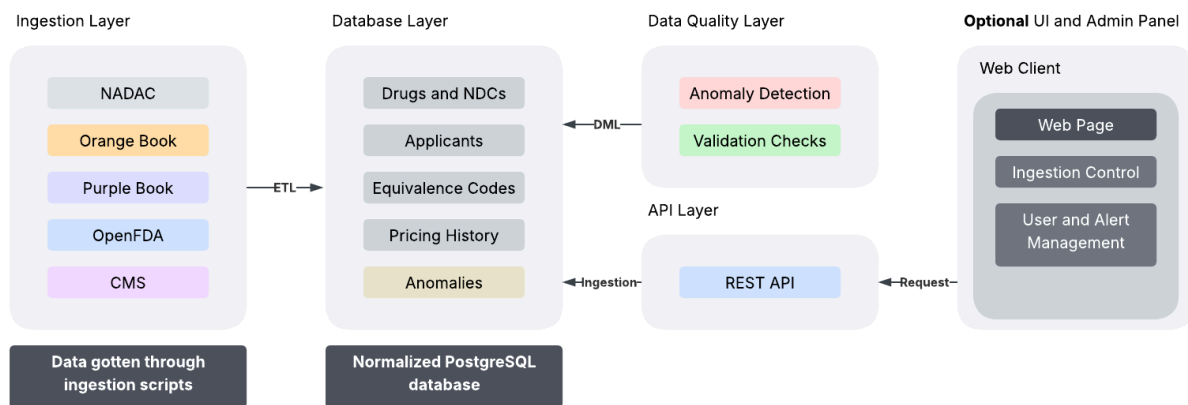
Proposed Solution

We propose a modular hub consisting of automated ingestion pipelines, a normalized PostgreSQL database, a data quality framework, and a REST API for internal querying. Optionally, we will design a lightweight web-based user interface and admin panel to enable non-technical users to explore the data and manage ingestion processes

Based on mentor guidance, we will use OpenFDA as the central integration layer. Using OpenFDA as a structured intermediary reduces ambiguity compared to name-based joins and improves reproducibility.

The database will be normalized to third normal form to eliminate redundancy and maintain referential integrity. Pricing tables will be time-indexed to support historical queries and volatility analysis. Data ingestion will occur weekly or via manual triggers, and the system will be deployed in Azure. We will also estimate required Azure credits based on expected storage and compute usage.

System Architecture



The architecture consists of five layers. First, ingestion scripts written in Python will fetch and preprocess public datasets. Second, transformed data will be loaded into a PostgreSQL database designed around normalized entities such as drugs, NDC-level products, applicants, therapeutic equivalence codes, and pricing records. Third, a data quality layer will flag anomalies, such as extreme price changes or missing mappings. Fourth, a REST API will allow secure querying by NDC, returning pricing information, package details, and equivalence data. Finally, an optional web-based UI and admin panel may be implemented to provide interactive search, visualization of price history, and administrative control over ingestion triggers and monitoring. This structure ensures modularity: ingestion, storage, validation, access, and presentation are clearly separated.

Novelty and Contribution

Our approach differs from existing practical implementations in several key ways. First, we explicitly design the system around a normalization strategy, rather than early denormalization. This makes the database extensible and analytically robust.

Second, our OpenFDA-centered mapping strategy provides a principled integration method that reduces cross-dataset ambiguity. Third, we incorporate built-in historical tracking, enabling time-traveling queries and volatility analysis. Finally, we implement automated anomaly detection rules to improve data reliability.

The optional UI and administrative layer further differentiates the system from purely backend integrations by transforming the infrastructure into an accessible internal data product. Together, these design decisions create not just a data repository, but a robust economic data infrastructure.

Intended Audience

The primary audience includes healthcare analytics teams, pharmaceutical market researchers, and internal data scientists. The system may also support billing and reimbursement teams if CMS crosswalk functionality is implemented. In future iterations, the infrastructure could power generic substitution engines or price risk forecasting tools.

If a UI layer is implemented, the system could also support non-technical stakeholders such as policy analysts or healthcare administrators who require direct access to pricing and equivalence data without interacting with raw APIs or SQL queries.

Preliminary Work

To date, we have met regularly with mentors to refine project scope and integration strategy. We identified OpenFDA as the central mapping layer and studied the structure of NDC codes to ensure accurate linking. We developed a working Jupyter notebook that successfully connects NADAC, Orange Book, Purple Book, and OpenFDA, validating that cross-source mappings through NDC and applicant number are feasible. This notebook serves as proof of concept for our ingestion and transformation logic.

The next phase involves designing the production database schema, deploying infrastructure in Azure, converting notebook logic into modular ingestion pipelines, and building the REST API.

Assumptions and Learning Goals

To keep the project feasible within the course timeline, we assume that public datasets remain accessible and that weekly refreshes are sufficient. Forecasting models, if implemented, will remain simple (e.g., ARIMA or Prophet).

Through this project, we aim to deepen our understanding of large-scale data engineering, schema normalization in real-world systems, handling schema drift, and designing infrastructure that supports both operational queries and data science use cases. More broadly, we want to understand how data infrastructure can improve transparency and economic decision-making in healthcare.

We also want to learn how to develop and evaluate a pharmaceutical pricing model built on normalized, time-indexed data. Through comparative experimentation with statistical and financial forecasting techniques, we aim to understand how different modeling assumptions impact predictive stability under conditions such as patent expiration, regulatory shocks, and gradual price inflation.